

TUM Hackathon Challenge

Automated Traffic Forecasting for Alpine Holiday Corridors A 8 East & A 93 South

Host

Die Autobahn GmbH des Bundes

1. Background & User Context

The Alpine motorway corridors **A 8 East and A 93 South** are critical routes for holiday and leisure traffic. Traffic volumes vary significantly depending on holidays, weather, and special events, making trip planning challenging for many user groups.

To support road users, Die Autobahn GmbH has developed a Traffic **Calendar** (“Fahrkalender”) that provides a **forward-looking traffic assessment for up to two months**, using a clear color-coded system from free-flowing traffic (green) to very heavy traffic (dark red).

The challenge:

Today, this calendar is largely based on **manual expert judgment**. Your task is to help transform this expert-driven approach into a scalable, data-driven forecasting system.

2. Created Value – User-Centric Perspective

Your solution should maximize value for real users who rely on predictable travel conditions.

Core User Stories

1. Private Traveler / Tourist

> As a holiday traveler, I want to know which days and time windows will have low traffic, so that I can plan my departure to avoid congestion and reduce travel stress.

2. Local Resident

> As a resident living near the A 8 / A 93 corridor, I want to know when very heavy traffic is expected, so that I can adapt my daily routines or avoid peak periods.

3. Logistics / Freight / Bus Company

> As a dispatcher, I want reliable traffic forecasts for specific days, so that I can schedule transports efficiently, inform customers proactively and avoid delays caused by holiday traffic.

4. Tourism Business

> As a hotel or tourism provider, I want advance insights into peak travel days, so that I can better coordinate arrivals, departures, and staffing.

5. Traffic Management Authority

> As a traffic planner, I want consistent and explainable forecasts, so that I can communicate traffic expectations transparently and support operational and project decisions.

3. Problem Statement

How can historical traffic data and publicly available information be used to **automatically generate daily, user-relevant traffic forecasts** that support early and informed planning for all these user groups?

4. Objective of the Challenge

Design and prototype a system that:

- Automatically generates daily traffic forecasts
- For all days of 2026–2029
- For the motorway corridors A 8 East and A 93 South (both directions)
- With a focus on holiday and leisure traffic

The system should reduce reliance on manual expert judgment while preserving the practical usability and clarity of the current Traffic Calendar.

5. Expected Outputs (from a User Perspective)

At minimum, your solution should provide, per day slots:

- Estimated traffic volume (e.g. vehicles per day)
- Typical daily traffic pattern (e.g. morning peak, afternoon peak)

Optional (bonus, user value):

- Clear traffic intensity categories (comparable to green → dark red)
- Confidence or uncertainty indicators
- Automatic identification of critical peak days

6. Data You Can Use

Provided by Die Autobahn GmbH:

- Historical traffic data for A 8 and A 93

Publicly Available Data (Optional but Encouraged)

- Weather data
- School holiday calendars (Germany + neighboring countries)
- Public holidays
- Known special events (e.g. border traffic control / “Dosierung” days)

7. Success Criteria (User-Oriented)

Your solution will be evaluated based on:

- Plausibility from a user perspective (“Would I trust this forecast?”)
- Relevance of the forecast for planning decisions
- Transparency and explainability
- Practical applicability in a public-sector context
- Innovation and smart use of data

8. Why This Challenge Matters

By working on this challenge, you help:

- Make holiday travel more predictable and less stressful
- Support better planning for businesses and residents
- Strengthen data-driven public services
- Scale expert knowledge into a sustainable forecasting tool

Let's turn historical data into real planning value for millions of road users.